

Junjie Hu

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EDUCATION

09/2023-6/2026 **Central South University, Changsha, China**

M.Sc., Traffic Engineering

Advisor: Prof. Dr. Jaeyoung Lee

09/2019-6/2023 **Central South University, Changsha, China**

B.Sc. Logistics Engineering

Advisor: Prof. Dr. Guohua Wu

PEER-REVIEWED JOURNAL PUBLICATIONS

- #1 **Hu, J.**, Hu, C., Yang, J., Bai, J., & Lee, J. J. (2024). Do traffic flow states follow Markov properties? A high-order spatiotemporal traffic state reconstruction approach for traffic prediction and imputation. *Chaos, Solitons, and Fractals*, 183(114965), 11,2024
- #2 **J. Hu**, J. Bai, J. Yang, and J. Lee, Crash Risk Prediction Using Sparse Collision Data: Causal Inference and Graph Convolutional Networks Approaches, *Expert Systems with Applications*, 125315,2025.
- #3 [Yang, J., Lee, J., Mao, S., & **Hu, J.**, Dynamic safety estimation of airport pick-up area based on video trajectory data, *IEEE Transactions on Intelligent Transportation Systems: A Publication of the IEEE Intelligent Transportation Systems Council*, 25\(2\), 1774–1786, 2024.](#)

SUBMISSION TO PEER-REVIEWED JOURNAL & CONFERENCES

- #1 **J. Hu**, Zhang, J. Bai, and J. Lee, Unraveling the Dynamic Rules of Urban Traffic Crashes: Tucker-Net Based SIRS Model, Available at SSRN: <https://ssrn.com/abstract=4697054> or <http://dx.doi.org/10.2139/ssrn.4697054>.
- #2 **J. Hu**, J. Lee, Car following dynamics in mixed traffic flow of autonomous and human-driven vehicles: complex networks approach, Under review
- #3 **J. Hu**, J. Bai, and J. Lee, Simplified and Efficient KNN-Based Method for High-Resolution Traffic Time Space Diagram Imputation, Under review.
- 2023 **World Transport Convention, Wuhan, China**
Presented a paper titled “Prediction of Traffic Risk based on Domain Transfer Graph Neural Network under Causal Inference”

AWARDS AND SCHOLARSHIPS

- 2024 2024 National Scholarship for Graduate Students
- 2023 China Graduate Student Mathematical Modelling Competition (**Second Prize**)
- 2022 The Mathematical Contest in Modeling Competition (**H Prize**)

2022	National College Students' Energy Conservation and Emission Reduction Competition (Third Prize)
2021	National College Student Logistics Simulation Competition (Third Prize)
2021	Asia and Pacific Mathematical Contest in Modeling (Third Prize)
2021	Chinese College Student Mathematical Modeling Competition (Second Prize)
2020-2022	Academic Scholarship for Central South University (Third Prize)

PROJECTS PARTICIPATED

2022-Present	Safety \& Planning Investigation and Research in Transportation (SPIRIT) Team Mentor: Prof. Jaeyoung Jay Lee My major responsibilities have encompassed proposing innovative ideas, designing and conducting research, composing research articles for publication, collaborating with fellow team members, and preparing to present research findings.
2023	Driving Path Guidance System Based on City-Level Trajectory With Safety Considerations Mentor: Prof. Ye Li I have completed the Classification of Unsafe Driving Behaviors and Constructing a Road Risk Assessment Model Based on Trajectory Data, the project was approved as an innovative industry project for college students in Hunan Province.
2023	Research on Collaborative Optimization and Incentive Mechanisms of Agricultural Waste Supply Chain Networks under the Rural Revitalization Strateg Mentor: Prof. Dezhi Zhang I have completed the investigation of biomass recycling in some areas of Hunan Province, and designed the collection and transportation network structure, and obtained the Third Prize in the National College Students' Energy Conservation and Emission Reduction Competition as the Project Leader.
2021	Intelligent Optimization Methods Personnel Mentor: Prof. Guohua Wu I am systematically trained in intelligent optimization methods and deep learning.

EXPERTISE

Programming Language: Proficient in: Python; Familiar with: Matlab, SPSS, Latex. Software: SPSS, GIS tools, Auto CAD, Mendeley. Languages: English, Chinese